

OSTMARK, Sweden

WMO: 024000

Lat: 60.350N

Lon: 12.650E

Elev: 1017

StdP: 14.16

Time Zone: 1.00 (EUC)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.6	(c) -0.2	(d) -10.4	(e) 3.3	(f) -5.4	(g) -4.7	(h) 4.4	(i) 0.3	(j) 34.4	(k) 37.6	(l) 33.0	(m) 37.5	(n) 1.6	(o) 320	(p) 0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.6	(c) 77.7	(d) 61.4	(e) 74.5	(f) 59.5	(g) 71.2	(h) 57.7	(i) 63.3	(j) 74.0	(k) 61.3	(l) 71.0	(m) 59.4	(n) 68.0	(o) 8.8	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.6	79.0	65.1	57.4	72.9	64.2	55.3	67.5	61.8	29.2	73.6	27.7	71.5	26.4	68.0	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.5	(b) 22.7	(c) 16.7	(e) -8.6	(f) 80.9	(g) 9.9	(h) 4.1	(i) -15.8	(j) 83.8	(k) -21.6	(l) 86.2	(m) -27.2	(n) 88.5	(o) -34.4	(p) 91.5		
(4) 26.5	(5) 22.7	(6) 16.7	(7) -8.6	(8) 80.9	(9) 9.9	(10) 4.1	(11) -15.8	(12) 83.8	(13) -21.6	(14) 86.2	(15) -27.2	(16) 88.5	(17) -34.4	(18) 91.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	23.0	23.8	29.9	36.6	46.8	54.2	59.5	55.3	47.6	38.5	29.6	23.3
	DBStd	14.85	11.29	10.34	7.53	6.48	6.76	5.77	5.22	5.22	5.52	6.68	7.66	9.43
	HDD50	4680	838	732	622	406	145	20	1	11	108	357	611	829
	HDD65	9475	1303	1152	1087	853	564	327	186	305	522	821	1061	1294
	CDD50	700	0	0	0	3	46	146	294	174	36	1	0	0
	CDD65	20	0	0	0	0	0	3	14	3	0	0	0	0
	CDH74	315	0	0	0	0	8	86	187	34	0	0	0	0
	CDH80	30	0	0	0	0	0	12	18	0	0	0	0	0
Wind	WSAvg	6.8	7.4	7.5	7.5	6.8	7.2	6.9	6.7	6.0	6.4	6.4	5.8	6.5
Precipitation	PrecAvg	32.9	2.3	1.7	1.5	1.9	2.5	3.5	3.7	3.9	3.0	3.3	3.1	2.6
	PrecMax	46.6	5.5	3.2	3.5	4.5	7.2	5.8	8.3	6.5	6.5	8.7	8.9	4.4
	PrecMin	26.9	0.5	0.5	0.2	0.4	0.2	0.3	1.4	1.2	0.8	0.9	1.2	0.4
	PrecStd	4.4	1.1	0.7	0.8	1.1	1.5	1.3	1.5	1.5	1.3	1.6	1.7	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.5	45.5	50.9	64.5	75.2	81.1	83.5	77.5	69.4	55.8	49.0	42.0
		MCWB	38.0	40.6	42.0	51.0	56.8	63.5	63.9	60.3	58.2	50.4	45.6	39.0
	2%	DB	39.9	42.1	45.4	57.1	71.5	75.9	79.2	74.5	63.3	52.1	45.9	39.6
		MCWB	36.4	38.5	38.0	46.1	55.2	60.3	62.4	59.8	53.9	47.5	42.5	37.6
	5%	DB	37.7	39.4	41.9	52.0	67.3	71.9	75.7	70.1	60.0	50.2	42.5	37.6
		MCWB	34.9	36.6	36.3	43.0	53.1	57.1	61.0	57.9	52.2	46.4	40.5	35.8
	10%	DB	35.6	36.9	39.2	48.0	61.8	67.6	71.9	65.9	57.1	48.3	40.2	35.3
		MCWB	33.4	34.4	35.1	40.8	50.1	55.1	59.0	56.4	50.9	44.9	38.7	33.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	41.4	43.6	51.9	58.4	64.4	66.2	63.8	60.3	52.0	46.3	40.1
		MCDB	41.1	43.7	48.8	62.8	71.5	78.5	78.8	71.0	65.4	54.3	48.6	41.2
	2%	WB	37.4	39.3	39.6	47.2	56.1	61.7	64.1	61.7	56.6	49.3	43.0	38.1
		MCDB	39.2	41.2	42.6	54.1	69.6	73.8	75.3	69.1	60.6	50.9	44.9	39.4
	5%	WB	35.5	37.1	37.6	44.4	53.6	58.5	62.4	59.9	54.3	47.5	40.9	36.2
		MCDB	37.1	39.1	40.2	51.2	66.0	69.0	73.4	67.7	58.1	49.2	42.2	37.2
	10%	WB	33.6	34.6	35.8	41.7	50.9	56.2	60.2	57.7	51.8	45.6	38.9	33.9
		MCDB	35.1	36.3	39.0	46.4	61.3	65.5	69.7	64.5	55.5	47.6	40.3	35.0
Mean Daily Temperature Range	5% DB	MDBR	9.3	11.6	12.9	15.8	20.4	18.1	19.6	16.6	15.6	10.3	8.2	9.6
		MCDBR	9.8	9.5	15.0	23.3	29.4	27.8	25.3	23.2	20.3	12.3	10.0	10.1
		MCWBR	9.4	8.2	9.9	13.2	14.8	13.4	11.6	9.5	11.1	9.0	8.9	9.8
	5% WB	MCDBR	9.1	9.3	12.4	21.9	28.4	25.1	22.3	18.3	14.6	9.3	9.4	9.7
		MCWBR	9.2	8.8	9.1	12.7	15.1	13.0	11.4	9.4	8.7	8.2	8.8	9.5
Clear-Sky Solar Irradiance	taub	0.242	0.286	0.312	0.350	0.351	0.342	0.362	0.353	0.323	0.310	0.262	0.215	
	taud	2.208	2.306	2.356	2.378	2.439	2.484	2.441	2.462	2.536	2.488	2.291	2.103	
	Ebn at Noon	173	229	260	269	278	282	272	266	257	220	163	130	
	Edh at Noon	12	20	27	31	32	31	32	29	23	17	11	8	
All-Sky Solar Radiation	RadAvg	89	294	768	1211	1506	1718	1532	1202	799	357	113	56	
	RadStd	14	56	85	152	148	108	212	126	98	70	32	7	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page